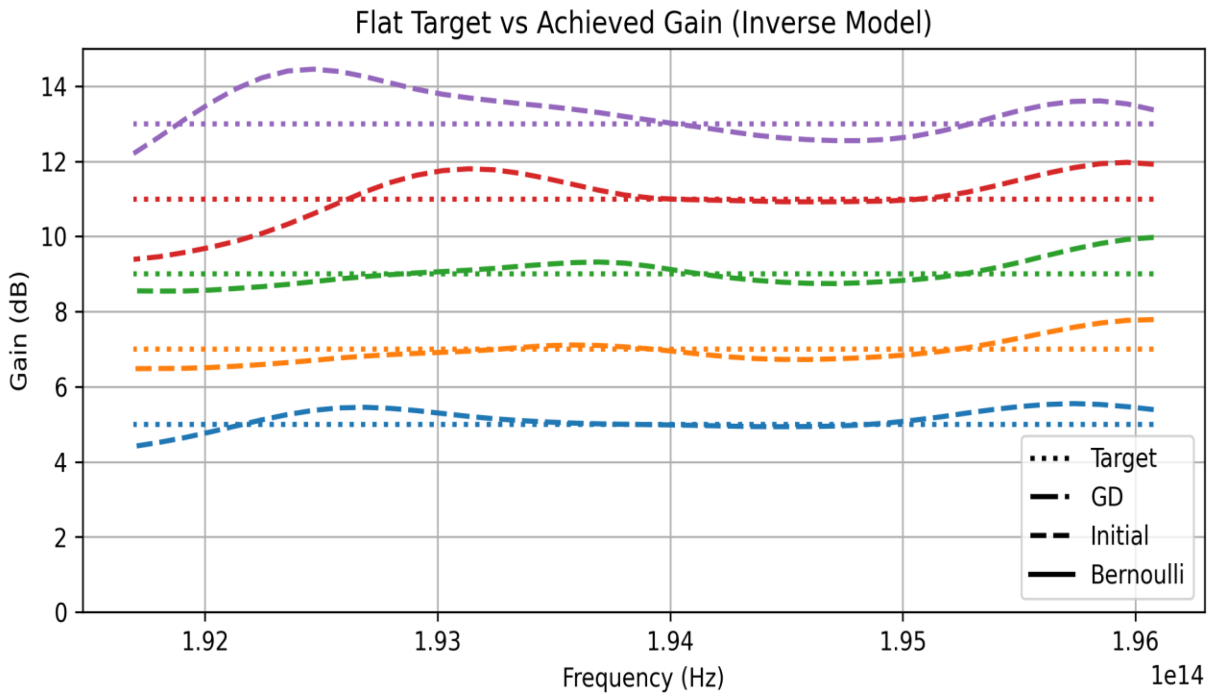
DTU

Reinforcement Learning Control of Raman Amplifiers 15.01.2026

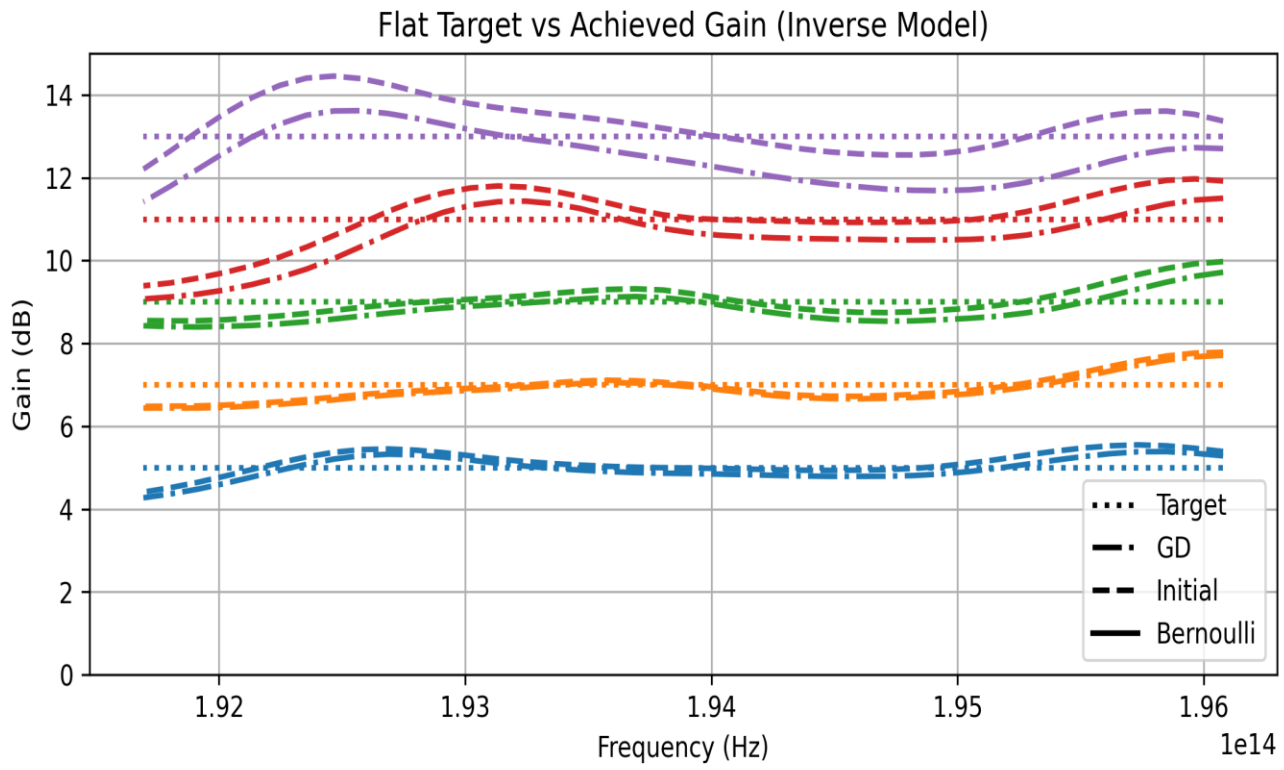
Inverse model

- The inverse model uses Random Projection Method/Extreme Learning to capture the inverse mapping of the non-linear system.
- It takes the output spectrum as input, and outputs the powers and wavelengths needed to achieve the desired spectrum.
- Experimentation with the hyper-parameters resulted in performance of the inverse model
- It achieves the error of:
  - Mean: 0.471 dB
  - Std: 0.308 dB
- This error statistic is close to the one achieved in the JLT paper (mean: 0.485 dB, std: 0.262 dB)



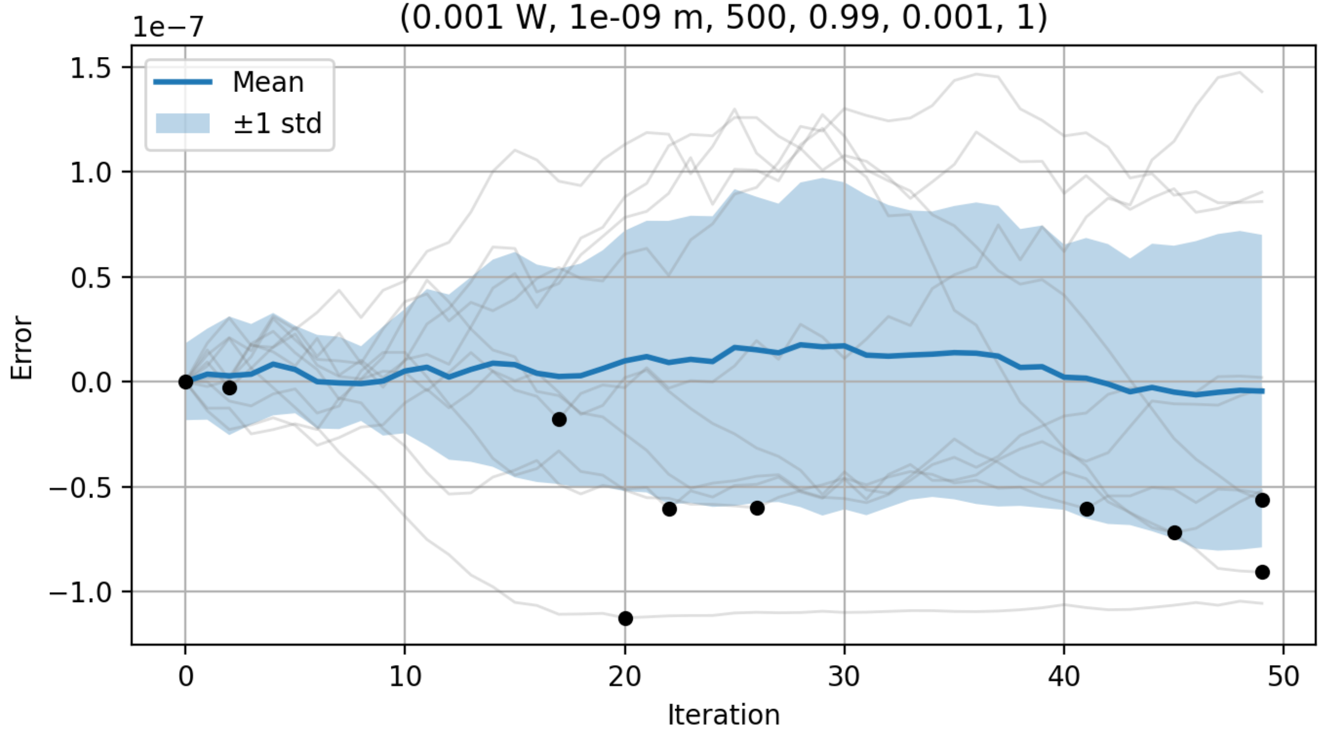
Inverse model + Gradient Descent Optimization

- Using the gradient descent based on the forward model's gradient reduces the error.
- The stopping criterium
  used was running the gradient descent optimization until the error stared increasing.
- Gradient descent works well in this case since the inverse model provides a good starting point which is close to the desired minimum

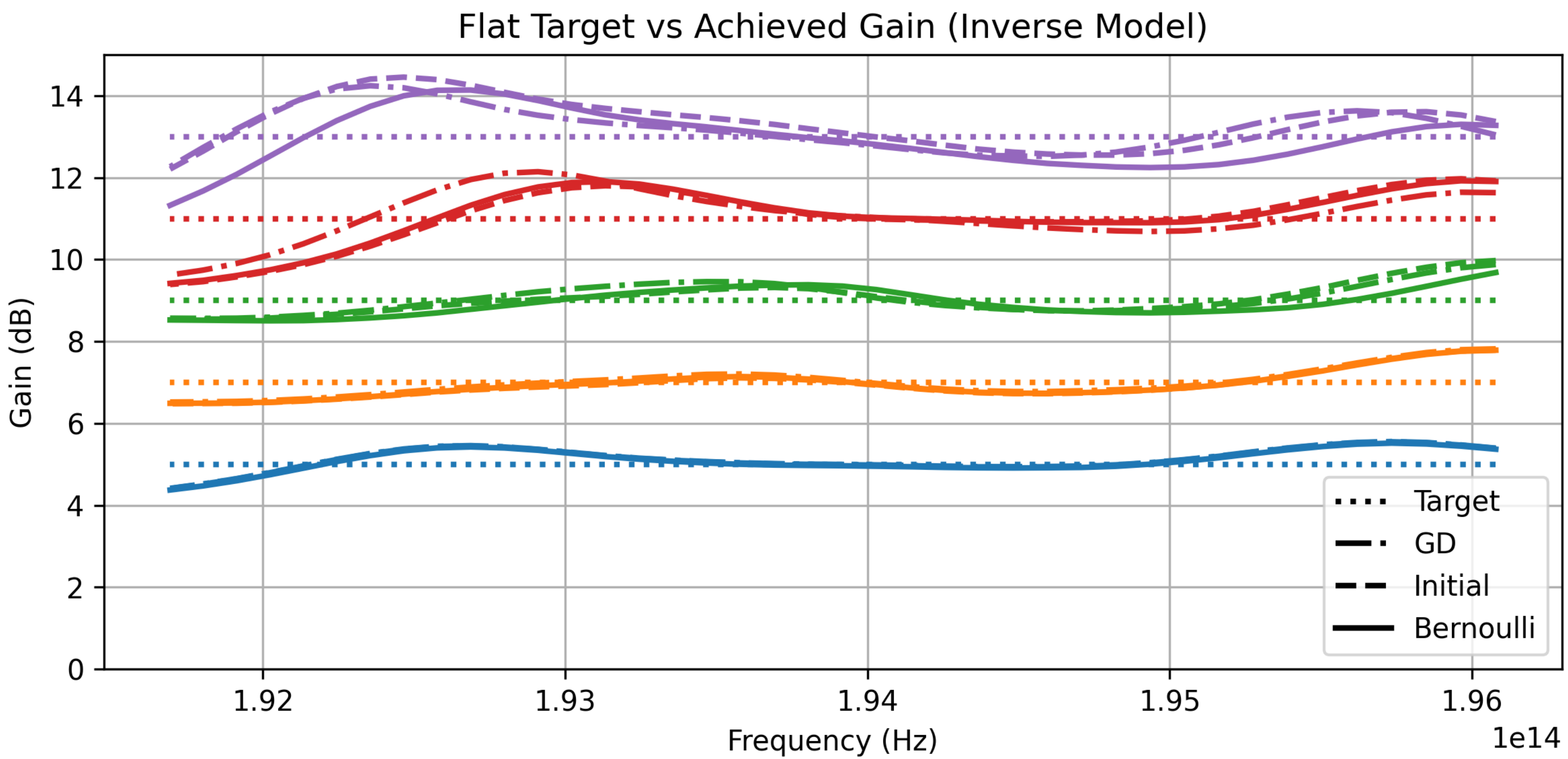


RL Controller – MSE Loss

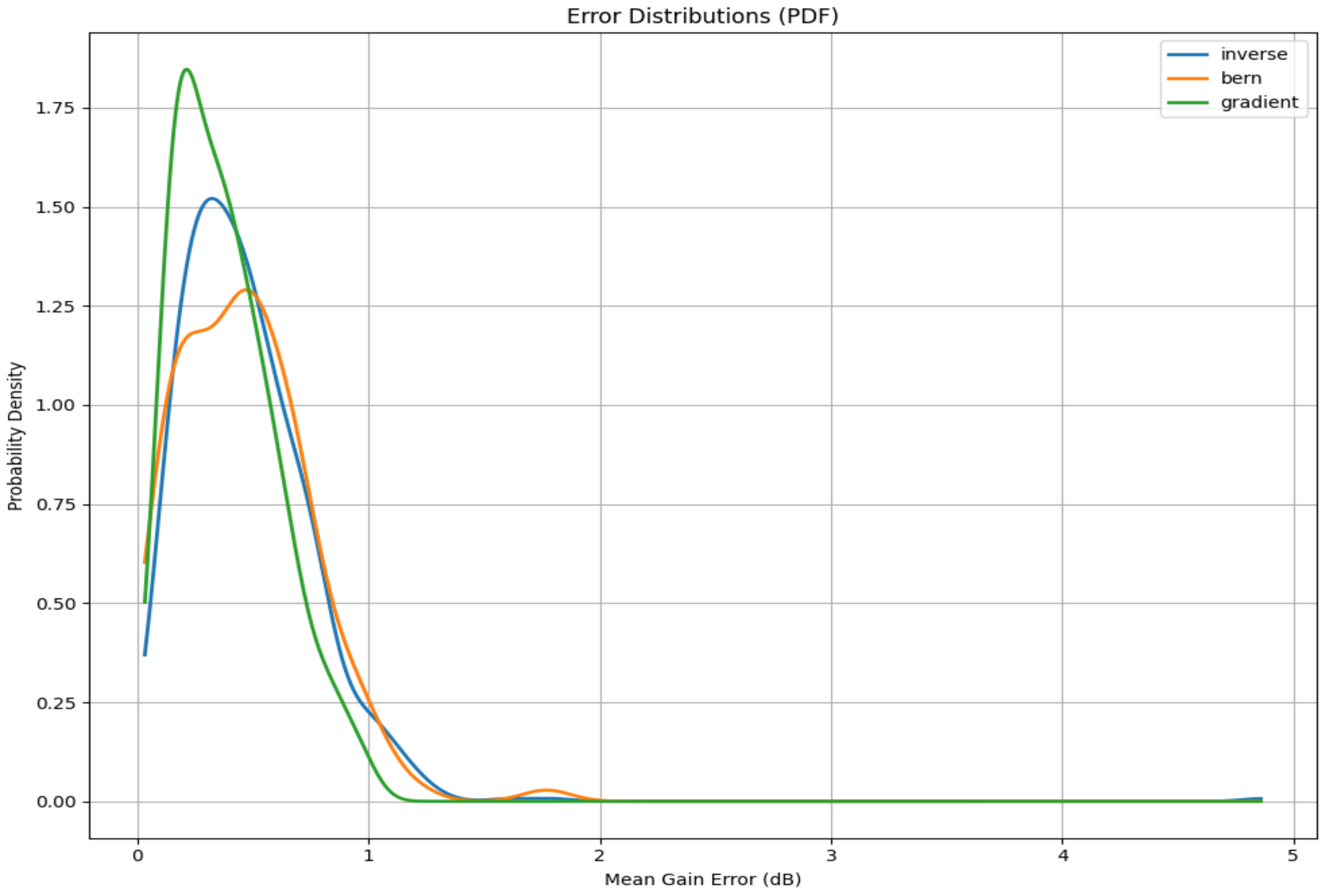
- Hyper-parameter tuning of Bernoulli controller didn't result in better performance so far.
- Optimizing hyper-parameters for much simpler problems proved quite difficult as well, so it could be a matter of persistence.
- Alternatively, since the Bernoulli controller does a good job at exploring the input space, it could use memory of the best set of parameters, which could be reproduced upon the end of optimization.



Inverse model + RL Optimization



Inverse Model vs Gradient Descent vs Bernoulli Controller MSE Statistics



- Inverse model:
  - N = 711
  - μ = 0.471 dB
  - σ = 0.308 dB
- Gradient Descent:
  - N = 167
  - μ = 0.382 dB
  - σ = 0.213 dB
- Bernoulli Controller:
  - N = 133
  - μ = 0.463 dB
  - σ = 0.281 dB